

scanFiles

June 13, 2025

```
[116]: import os
import re
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

[95]: def count_words_in_go_files(folder_path, words_dict):
    word_search_counts = {word: 0 for word in words_dict.keys()}
    word_markup_counts = words_dict.copy()

    pattern = re.compile(r'\b(' + '|'.join(map(re.escape, words_dict.keys())) +
↳r')\b')

    for root, dirs, files in os.walk(folder_path):
        for file in files:
            if file.endswith('.go'):
                file_path = os.path.join(root, file)
                try:
                    with open(file_path, 'r', encoding='utf-8') as f:
                        content = f.read()
                        matches = pattern.findall(content)
                        for word in matches:
                            word_search_counts[word] += 1
                except (UnicodeDecodeError, PermissionError) as e:
                    print(f"                {file_path}: {e}")

    # DataFrame
    df = pd.DataFrame({
        'Word': list(word_search_counts.keys()),
        'Count Search': list(word_search_counts.values()),
        'Count Markup': [word_markup_counts[word] for word in
↳word_search_counts.keys()]
    })

    # Ratio
    df['Ratio'] = df['Count Search'] / df['Count Markup']
```

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#         Ratio (      )
df_sorted = df.sort_values(by='Ratio', ascending=False)
df_sorted.reset_index(drop=True, inplace=True)

return df_sorted

```

```
[96]: global_array = np.array([])
```

```

[97]: def analyze(folder, words):
    result_df = count_words_in_go_files(folder, words)

    mean_row = pd.DataFrame({
        'Word': ['MEAN'],
        'Count Search': [result_df['Count Search'].mean()],
        'Count Markup': [result_df['Count Markup'].mean()],
        'Ratio': [result_df['Ratio'].mean()]
    })

    global global_array
    global_array = np.append(global_array, result_df['Ratio'].values)

    display(result_df)
    display(mean_row)

```

```

[98]: folder = 'e:\\phd\\test_repos\\backend\\fsm-gateway\\'
words = {
    "GetOrderById": 1,
    "GetSession": 1,
    "GetDocumentData": 1,
    "GetDocumentText": 1,
    "GetConsentList": 1,
    "StartSession": 1,
    "SendDocumentData": 1,
    "FinishSession": 1,
    "Logout": 1,
    "StartAuthSessionByPhone": 1,
    "GetJwtByPhoneCode": 1,
    "GetJwtByRefreshToken": 1
}
analyze(folder, words)

```

	Word	Count Search	Count Markup	Ratio
0	GetJwtByRefreshToken	25	1	25.0
1	StartAuthSessionByPhone	21	1	21.0
2	GetJwtByPhoneCode	21	1	21.0
3	GetSession	11	1	11.0
4	GetDocumentData	10	1	10.0
5	StartSession	10	1	10.0

6	GetDocumentText	9	1	9.0
7	GetConsentList	9	1	9.0
8	SendDocumentData	9	1	9.0
9	FinishSession	9	1	9.0
10	GetOrderById	4	1	4.0
11	Logout	1	1	1.0

	Word	Count	Search	Count	Markup	Ratio
0	MEAN		11.583333		1.0	11.583333

```
[99]: folder = 'e:\\phd\\test_repos\\LP\\delivery-offering\\'
words = {
    "DeliveryTimeslots": 2,
    "DeliveryTimeslot": 2,
    "DeliveryTypes": 2,
    "ReserveDeliveryTimeslot": 2,
    "LookupDeliveryOffers": 2,
    "DeliveryOfferByID": 2,
    "AcceptDeliveryOffer": 2,
    "ValidateDeliveryOffer": 2,
}
analyze(folder, words)
```

	Word	Count	Search	Count	Markup	Ratio
0	DeliveryTimeslot		94		2	47.0
1	DeliveryTypes		29		2	14.5
2	LookupDeliveryOffers		22		2	11.0
3	AcceptDeliveryOffer		19		2	9.5
4	DeliveryOfferByID		17		2	8.5
5	ReserveDeliveryTimeslot		16		2	8.0
6	DeliveryTimeslots		12		2	6.0
7	ValidateDeliveryOffer		12		2	6.0

	Word	Count	Search	Count	Markup	Ratio
0	MEAN		27.625		2.0	13.8125

```
[ ]:
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```
[100]: folder = 'e:\\phd\\test_repos\\LP\\warehouses\\'
words = {
    "TerminateWarehouses": 2,
    "UpdateWarehouses": 2,
    "CreateWarehouses": 2,
    "Warehouses": 2,
}
analyze(folder, words)
```

	Word	Count	Search	Count	Markup	Ratio
0	Warehouses		71		2	35.5
1	UpdateWarehouses		23		2	11.5

2	CreateWarehouses	22	2	11.0
3	TerminateWarehouses	20	2	10.0
	Word	Count	Search	Count Markup Ratio
0	MEAN	34.0	2.0	17.0

```
[101]: folder = 'e:\\phd\\test_repos\\LP\\channel-profile\\'
words = {
    "CreateChannelProfile": 2,
    "UpdateChannelProfile": 2,
    "DeleteChannelProfile": 2,
    "ChannelProfiles": 2,
    "ChannelsByFilter": 2,
}
analyze(folder, words)
```

	Word	Count	Search	Count Markup	Ratio
0	ChannelsByFilter	17	2	8.5	
1	CreateChannelProfile	16	2	8.0	
2	UpdateChannelProfile	16	2	8.0	
3	DeleteChannelProfile	16	2	8.0	
4	ChannelProfiles	15	2	7.5	
	Word	Count	Search	Count Markup	Ratio
0	MEAN	16.0	2.0	8.0	

```
[102]: folder = 'e:\\phd\\test_repos\\modules\\'
words = {
    "CompleteFollowup": 1,
    "ReAssignFollowup": 2,
    "RemovePharmacy": 1,
    "SetABVariant": 1,
    "BlockPatient": 1,
    "SetMediaRead": 1,
    "ValidateAppointmentInsurance": 3,
    "UpdateProviderInsurance": 1,
    "EditStateCompanies": 1,
    "UpdateEncounterScreenedStatus": 1,
    "StartPaymentTransaction": 1,
    "SubscribeToPhoneCallEvents": 1,
    "SetAppointmentType": 1,
    "MakeCardDefault": 2,
    "DeletePatientNote": 1,
    "UpdatePatientNote": 1,
    "SetVisitTypeOrders": 2,
    "SetVisitTypeVisibility": 2,
    "AddICD10MedicalCode": 2,
    "AddMedicalCodes": 1,
    "InviteClient": 1,
}
```

```

"DeleteEvent": 1,
"NeedFollowup": 1,
"TwilioInfo": 1,
"CurrentTime": 1,
"Person": 1,
"PHIDData": 1,
"Clients": 1,
"Messages": 1,
"PastVisits": 1,
"Appointment": 1,
"CheckInData": 1,
"NoteTemplate": 1,
"VisitType": 1,
"ScheduleVisit": 1,
"UpdateAvailability": 1,
"AssignQuestionnaire": 1,
"PublishForm": 1,
"LoadFollowupByPatients": 1,
"InsuranceCosts": 1,
"PatientDetails": 1,
"ConsentForms": 1
}
analyze(folder, words)

```

	Word	Count	Search	Count	Markup	Ratio
0	Appointment		219		1	219.000000
1	VisitType		123		1	123.000000
2	Person		111		1	111.000000
3	Messages		35		1	35.000000
4	PHIDData		27		1	27.000000
5	NoteTemplate		26		1	26.000000
6	CheckInData		12		1	12.000000
7	PatientDetails		11		1	11.000000
8	TwilioInfo		8		1	8.000000
9	DeletePatientNote		7		1	7.000000
10	StartPaymentTransaction		7		1	7.000000
11	UpdatePatientNote		7		1	7.000000
12	CurrentTime		6		1	6.000000
13	UpdateAvailability		5		1	5.000000
14	NeedFollowup		5		1	5.000000
15	Clients		5		1	5.000000
16	SetAppointmentType		5		1	5.000000
17	MakeCardDefault		9		2	4.500000
18	AddICD10MedicalCode		8		2	4.000000
19	RemovePharmacy		4		1	4.000000
20	BlockPatient		4		1	4.000000
21	SubscribeToPhoneCallEvents		4		1	4.000000
22	EditStateCompanies		4		1	4.000000

23	UpdateProviderInsurance	4	1	4.000000
24	SetABVariant	4	1	4.000000
25	ValidateAppointmentInsurance	11	3	3.666667
26	SetVisitTypeVisibility	7	2	3.500000
27	ReAssignFollowup	7	2	3.500000
28	SetVisitTypeOrders	7	2	3.500000
29	PublishForm	2	1	2.000000
30	AssignQuestionnaire	2	1	2.000000
31	LoadFollowupByPatients	1	1	1.000000
32	InsuranceCosts	1	1	1.000000
33	CompleteFollowup	1	1	1.000000
34	DeleteEvent	1	1	1.000000
35	ScheduleVisit	1	1	1.000000
36	PastVisits	1	1	1.000000
37	InviteClient	1	1	1.000000
38	AddMedicalCodes	1	1	1.000000
39	UpdateEncounterScreenedStatus	1	1	1.000000
40	SetMediaRead	1	1	1.000000
41	ConsentForms	1	1	1.000000

	Word	Count	Search	Count	Markup	Ratio
0	MEAN	16.833333		1.166667	16.206349	

[]:

```
[103]: folder = 'e:\\phd\\test_repos_graphql2\\boost\\'
words = {
```

```
    "DealUpdate": 1,
    "StorageAskUpdate": 1,
    "FundsMoveToEscrow": 1,
    "DealFailPaused": 1,
    "DealRetryPaused": 1,
    "DealCancel": 1,
    "Mpool": 1,
    "FundsLogs": 1,
    "ProposalLogs": 1,
    "LegacyDeals": 1,
    "Deals": 1,
    "LegacyDeal": 1,
    "Deal": 2,
    "DealNew": 1,
    "DealsCount": 1,
    "StorageAsk": 1,
    "DealPublishNow": 1,
    "DealPublish": 1,
    "Epoch": 1,
    "Funds": 1,
    "ProposalLogsCount": 1,
```

```

    "SealingPipeline": 1,
    "Storage": 1,
    "Transfers": 1,
}
analyze(folder, words)

```

	Word	Count	Search	Count	Markup	Ratio
0	Deals		56		1	56.0
1	StorageAsk		32		1	32.0
2	Storage		26		1	26.0
3	Deal		16		2	8.0
4	Epoch		7		1	7.0
5	ProposalLogs		5		1	5.0
6	FundsLogs		4		1	4.0
7	Transfers		4		1	4.0
8	DealNew		2		1	2.0
9	DealPublish		2		1	2.0
10	Funds		2		1	2.0
11	SealingPipeline		2		1	2.0
12	Mpool		1		1	1.0
13	LegacyDeals		1		1	1.0
14	DealCancel		1		1	1.0
15	LegacyDeal		1		1	1.0
16	StorageAskUpdate		1		1	1.0
17	DealsCount		1		1	1.0
18	DealRetryPaused		1		1	1.0
19	DealPublishNow		1		1	1.0
20	DealFailPaused		1		1	1.0
21	ProposalLogsCount		1		1	1.0
22	FundsMoveToEscrow		1		1	1.0
23	DealUpdate		1		1	1.0

	Word	Count	Search	Count	Markup	Ratio
0	MEAN		7.083333		1.041667	6.75

```

[104]: folder = 'e:\\phd\\test_repos_graphql2\\bramble\\'
words = {
    "UploadGadgetFile": 1,
    "UploadGizmoFile": 1,
    "Gizmo": 2,
    "Gizmos": 2,
    "Delay": 1,
    "Gizmos": 2,
    "RandomGizmo": 1,
}
analyze(folder, words)

```

	Word	Count	Search	Count	Markup	Ratio
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0	Gizmo	297	2	148.5
1	UploadGadgetFile	1	1	1.0
2	UploadGizmoFile	1	1	1.0
3	Gizmos	2	2	1.0
4	Delay	1	1	1.0
5	RandomGizmo	1	1	1.0

	Word	Count	Search	Count	Markup	Ratio
0	MEAN		50.5		1.333333	25.583333

```
[105]: folder = 'e:\\phd\\test_repos_graphql2\\circuit\\'
```

```
words = {
    "Secrets": 2,
    "Services": 2,
    "Releases": 2,
    "EnvsDeployedIn": 2,
    "Features": 2,
    "GetGitCommits": 1,
    "BookmarkProject": 1,
    "UpdateProjectEnvironments": 1,
    "UpdateUserPermissions": 1,
    "DeleteProjectExtension": 1,
    "UpdateProjectExtension": 1,
    "CreateProjectExtension": 1,
    "DeleteExtension": 1,
    "UpdateExtension": 1,
    "CreateExtension": 1,
    "ImportSecrets": 1,
    "DeleteSecret": 1,
    "UpdateSecret": 1,
    "CreateSecret": 1,
    "DeleteEnvironment": 1,
    "UpdateEnvironment": 1,
    "CreateEnvironment": 1,
    "DeleteServiceSpec": 1,
    "UpdateServiceSpec": 1,
    "CreateServiceSpec": 1,
    "ImportServices": 1,
    "DeleteService": 1,
    "UpdateService": 1,
    "CreateService": 1,
    "StopRelease": 1,
    "CreateRelease": 1,
    "UpdateProject": 1,
    "CreateProject": 1,
    "Permissions": 1,
    "ReleaseExtensions": 1,
    "ProjectExtensions": 1,
```

```

    "Extensions": 1,
    "Secret": 1,
    "ExportSecrets": 1,
    "Environments": 1,
    "ServiceSpecs": 1,
    "Users": 1,
    "User": 1,
    "Bookmarked": 1,
  }
  analyze(folder, words)

```

	Word	Count Search	Count Markup	Ratio
0	User	2577	1	2577.0
1	Secret	660	1	660.0
2	Extensions	379	1	379.0
3	Permissions	371	1	371.0
4	Services	541	2	270.5
5	CreateProject	180	1	180.0
6	CreateEnvironment	164	1	164.0
7	Secrets	299	2	149.5
8	CreateService	143	1	143.0
9	Users	122	1	122.0
10	CreateSecret	99	1	99.0
11	UpdateService	82	1	82.0
12	UpdateProject	66	1	66.0
13	CreateExtension	61	1	61.0
14	CreateServiceSpec	55	1	55.0
15	UpdateSecret	54	1	54.0
16	CreateProjectExtension	54	1	54.0
17	UpdateEnvironment	41	1	41.0
18	DeleteService	39	1	39.0
19	CreateRelease	37	1	37.0
20	Environments	33	1	33.0
21	Features	62	2	31.0
22	DeleteSecret	27	1	27.0
23	ProjectExtensions	26	1	26.0
24	DeleteEnvironment	21	1	21.0
25	Bookmarked	17	1	17.0
26	ImportServices	12	1	12.0
27	ReleaseExtensions	12	1	12.0
28	StopRelease	12	1	12.0
29	ImportSecrets	10	1	10.0
30	UpdateProjectEnvironments	10	1	10.0
31	Releases	19	2	9.5
32	UpdateUserPermissions	9	1	9.0
33	DeleteServiceSpec	9	1	9.0
34	DeleteProjectExtension	8	1	8.0
35	UpdateProjectExtension	8	1	8.0

36	UpdateServiceSpec	8	1	8.0
37	BookmarkProject	8	1	8.0
38	DeleteExtension	7	1	7.0
39	ExportSecrets	7	1	7.0
40	UpdateExtension	7	1	7.0
41	ServiceSpecs	4	1	4.0
42	EnvsDeployedIn	8	2	4.0
43	GetGitCommits	3	1	3.0

	Word	Count	Search	Count	Markup	Ratio
0	MEAN	144.795455		1.113636	134.238636	

```
[106]: folder = 'e:\\phd\\test_repos_graphql2\\core-geth\\'
words = {
    "SendRawTransaction": 2,
    "Logs": 2,
    "Transaction": 2,
    "Blocks": 1,
    "Block": 1,
    "EstimateGas": 1,
    "Call": 1,
    "Account": 1,
    "EstimateGas": 1,
    "Call": 1,
    "Account": 1,
    "OmmerAt": 1,
    "Miner": 1,
    "CreatedContract": 1,
    "Account": 1,
    "Storage": 1,
    "Status": 1,
    "Nonce": 1,
    "Hash": 1,
    "TransactionCount": 1,
    "ChainID": 1,
    "MaxPriorityFeePerGas": 1,
    "GasPrice": 1,
    "Transactions": 1,
    "ExcessBlobGas": 1,
    "BlobGasUsed": 1,
    "Withdrawals": 1,
    "WithdrawalsRoot": 1,
    "Raw": 1,
    "RawHeader": 1,
    "TotalDifficulty": 1,
    "LogsBloom": 1,
    "ExtraData": 1,
    "Ommers": 1,
```

```

    "OmmerCount": 1,
    "OmmerHash": 1,
    "ReceiptsRoot": 1,
    "StateRoot": 1,
    "TransactionsRoot": 1,
    "MixHash": 1,
    "Timestamp": 1,
    "Difficulty": 1,
    "Parent": 1,
    "NextBaseFeePerGas": 1,
    "BaseFeePerGas": 1,
    "GasUsed": 1,
    "GasLimit": 1,
    "Number": 1,
    "YParity": 1,
    "AccessList": 1,
    "Type": 1,
    "BlobGasPrice": 1,
    "CumulativeGasUsed": 1,
    "Index": 1,
    "Value": 1,
    "EffectiveTip": 1,
    "BlobVersionedHashes": 1,
    "MaxFeePerBlobGas": 1,
    "MaxFeePerGas": 1,
    "EffectiveGasPrice": 1,
    "Gas": 1,
    "InputData": 1
}
analyze(folder, words)

```

	Word	Count	Search	Count	Markup	Ratio
0	Hash		6364		1	6364.0
1	Number		1446		1	1446.0
2	Value		1096		1	1096.0
3	Type		1049		1	1049.0
4	Block		770		1	770.0
5	Nonce		712		1	712.0
6	Account		442		1	442.0
7	Transaction		852		2	426.0
8	Gas		420		1	420.0
9	Difficulty		387		1	387.0
10	Transactions		345		1	345.0
11	GasLimit		341		1	341.0
12	GasPrice		319		1	319.0
13	Storage		253		1	253.0
14	ChainID		246		1	246.0
15	GasUsed		219		1	219.0

16	Timestamp	188	1	188.0
17	AccessList	182	1	182.0
18	Call	169	1	169.0
19	Status	168	1	168.0
20	Index	156	1	156.0
21	Withdrawals	148	1	148.0
22	ExcessBlobGas	146	1	146.0
23	BlobGasUsed	131	1	131.0
24	Miner	105	1	105.0
25	Logs	164	2	82.0
26	MaxFeePerGas	82	1	82.0
27	ExtraData	73	1	73.0
28	MaxPriorityFeePerGas	72	1	72.0
29	Parent	60	1	60.0
30	Blocks	59	1	59.0
31	CumulativeGasUsed	51	1	51.0
32	StateRoot	51	1	51.0
33	Raw	47	1	47.0
34	BaseFeePerGas	32	1	32.0
35	LogsBloom	23	1	23.0
36	EffectiveGasPrice	23	1	23.0
37	TotalDifficulty	18	1	18.0
38	BlobGasPrice	17	1	17.0
39	EstimateGas	17	1	17.0
40	BlobVersionedHashes	17	1	17.0
41	Ommers	17	1	17.0
42	MixHash	16	1	16.0
43	ReceiptsRoot	16	1	16.0
44	WithdrawalsRoot	14	1	14.0
45	YParity	12	1	12.0
46	OmmmerHash	12	1	12.0
47	TransactionCount	9	1	9.0
48	MaxFeePerBlobGas	8	1	8.0
49	TransactionsRoot	4	1	4.0
50	EffectiveTip	2	1	2.0
51	SendRawTransaction	4	2	2.0
52	NextBaseFeePerGas	2	1	2.0
53	OmmmerCount	2	1	2.0
54	RawHeader	2	1	2.0
55	CreatedContract	2	1	2.0
56	OmmmerAt	2	1	2.0
57	InputData	2	1	2.0

	Word	Count	Search	Count	Markup	Ratio
0	MEAN	303.206897		1.051724	294.413793	

```
[107]: folder = 'e:\\phd\\test_repos_graphql2\\exo\\'
words = {
```

```

    "Components": 2,
    "FindManifest": 1,
    "WatchJob": 1,
    "StackByID": 1,
    "DestroyComponents": 1,
    "UpdateComponent": 2,
    "DestroyStack": 1,
    "FinishTask": 1,
    "AcquireTask": 1,
    "DisposeResource": 1,
    "UpdateResource": 2,
    "RestartWorkspaceComponents": 1,
    "CreateWorkspace": 2,
    "CreateProject": 1,
    "ResourceByID": 1,
    "MakeManifest": 1,
    "FindEntity": 1,
    "Stack": 1,
    "Manifest": 1,
    "BuildWorkspace": 1,
    "DestroyWorkspace": 1,
    "TaskByID": 1,
    "DisplayName": 1,
    "ForgetResource": 1,
    "ReconcileComponent": 1,
    "InitializeResource": 1,
    "RefreshResource": 1,
    "RefreshCluster": 1,
    "SetWorkspaceStack": 1,
    "Build": 1,
    "Project": 1,
}
analyze(folder, words)

```

	Word	Count	Search	Count	Markup	Ratio
0	Stack		85		1	85.0
1	DisplayName		75		1	75.0
2	Build		54		1	54.0
3	Components		75		2	37.5
4	Project		36		1	36.0
5	Manifest		26		1	26.0
6	CreateProject		5		1	5.0
7	CreateWorkspace		7		2	3.5
8	DestroyComponents		3		1	3.0
9	UpdateResource		6		2	3.0
10	DestroyWorkspace		2		1	2.0
11	UpdateComponent		4		2	2.0
12	RestartWorkspaceComponents		1		1	1.0

13	StackByID	1	1	1.0
14	WatchJob	1	1	1.0
15	SetWorkspaceStack	1	1	1.0
16	RefreshCluster	1	1	1.0
17	RefreshResource	1	1	1.0
18	InitializeResource	1	1	1.0
19	ReconcileComponent	1	1	1.0
20	ForgetResource	1	1	1.0
21	TaskByID	1	1	1.0
22	DisposeResource	1	1	1.0
23	FinishTask	1	1	1.0
24	BuildWorkspace	1	1	1.0
25	DestroyStack	1	1	1.0
26	AcquireTask	1	1	1.0
27	FindEntity	1	1	1.0
28	FindManifest	1	1	1.0
29	ResourceByID	1	1	1.0
30	MakeManifest	1	1	1.0

	Word	Count	Search	Count	Markup	Ratio
0	MEAN		12.806452		1.129032	11.322581

```
[108]: folder = 'e:\\phd\\test_repos_graphql2\\go-graphql-starter\\'
words = {
    "Signup": 1,
    "User": 1,
    "Login": 1,
    "Viewer": 1,
}
analyze(folder, words)
```

	Word	Count	Search	Count	Markup	Ratio
0	User		3		1	3.0
1	Signup		1		1	1.0
2	Login		1		1	1.0
3	Viewer		1		1	1.0

	Word	Count	Search	Count	Markup	Ratio
0	MEAN		1.5		1.0	1.5

```
[109]: folder = 'e:\\phd\\test_repos_graphql2\\HydroAPI\\'
words = {
    "Merit": 1,
    "Strike": 1,
    "Giveaway": 1,
    "Server": 1,
    "User": 1,
    "Msgs": 1,
}
```

```
analyze(folder, words)
```

	Word	Count	Search	Count	Markup	Ratio
0	Server		109		1	109.0
1	User		95		1	95.0
2	Giveaway		54		1	54.0
3	Merit		44		1	44.0
4	Strike		44		1	44.0
5	Msgs		7		1	7.0

	Word	Count	Search	Count	Markup	Ratio
0	MEAN		58.833333		1.0	58.833333

```
[110]: folder = 'e:\\phd\\test_repos_graphql2\\mattermost-plugin-playbooks\\'
words = {
    "UpdateRunTaskActions": 1,
    "ChangeRunOwner": 1,
    "RemoveRunParticipants": 1,
    "AddRunParticipants": 1,
    "UpdateRun": 1,
    "SetRunFavorite": 1,
    "RemovePlaybookMember": 2,
    "AddPlaybookMember": 2,
    "DeleteMetric": 2,
    "UpdateMetric": 2,
    "AddMetric": 2,
    "UpdatePlaybook": 2,
    "UpdatePlaybookFavorite": 1,
    "Runs": 1,
    "Run": 1,
    "Playbooks": 1,
    "Playbook": 1
}
analyze(folder, words)
```

	Word	Count	Search	Count	Markup	Ratio
0	Run		463		1	463.0
1	Playbook		353		1	353.0
2	Playbooks		180		1	180.0
3	Runs		55		1	55.0
4	UpdateRunTaskActions		7		1	7.0
5	UpdatePlaybook		12		2	6.0
6	ChangeRunOwner		4		1	4.0
7	UpdateRun		4		1	4.0
8	UpdateMetric		7		2	3.5
9	AddMetric		7		2	3.5
10	DeleteMetric		7		2	3.5
11	SetRunFavorite		3		1	3.0
12	UpdatePlaybookFavorite		3		1	3.0

13	AddRunParticipants	3	1	3.0
14	RemoveRunParticipants	3	1	3.0
15	RemovePlaybookMember	5	2	2.5
16	AddPlaybookMember	5	2	2.5

	Word	Count	Search	Count	Markup	Ratio
0	MEAN		65.941176		1.352941	64.676471

```
[111]: folder = 'e:\\phd\\test_repos_graphql2\\SlabAPI\\'
words = {
    "GetViewCount": 2,
    "GetView": 1,
    "GetAllViews": 1,
    "GetUser": 1,
    "GetTrigger": 1,
    "GetAllTriggers": 1,
    "GetAllTickets": 1,
    "GetOrganization": 1,
}
analyze(folder, words)
```

	Word	Count	Search	Count	Markup	Ratio
0	GetTrigger		11		1	11.0
1	GetOrganization		11		1	11.0
2	GetView		10		1	10.0
3	GetUser		10		1	10.0
4	GetAllTriggers		7		1	7.0
5	GetAllTickets		7		1	7.0
6	GetAllViews		6		1	6.0
7	GetViewCount		10		2	5.0

	Word	Count	Search	Count	Markup	Ratio
0	MEAN		9.0		1.125	8.375

```
[112]: folder = 'e:\\phd\\test_repos_graphql2\\sturdy\\'
words = {
    "Integrations": 1,
    "UpdatedGitHubPullRequestStatuses": 1,
    "PullCodebase": 1,
    "CreateOrUpdateCodebaseRemote": 1,
    "UpdateInstallation": 1,
    "DeleteIntegration": 1,
    "SetupGitHubRepository": 1,
    "Activity": 1,
    "DownloadZip": 1,
    "Changes": 1,
    "Members": 2,
    "LastUsedView": 2,
    "UpdatedOrganization": 1,
}
```

```

    "UpdatedSuggestion": 1,
    "UpdatedView": 1,
    "UpdatedWorkspaceDiffs": 1,
    "GenerateKeyPair": 1,
    "RemovePatches": 2,
    "DismissSuggestionHunks": 1,
    "AddPublicKey": 1,
    "WatchWorkspace": 1,
    "RequestReview": 1,
    "UpdateACL": 1,
    "ArchiveNotifications": 1,
    "AddUserToCodebase": 1,
    "VerifyEmail": 2,
    "UpdateUser": 2,
    "DeleteComment": 1,
    "LandWorkspaceChange": 2,
    "Comment": 3,
    "Workspaces": 2,
    "ResolvedBy": 1,
    "Notifications": 1,
    "UpdateComment": 1,
    "CreateComment": 1,
    "UpdateCodebase": 1,
    "ReportWorkspacePresence": 1,
    "DismissSuggestion": 1,
    "UpdateOrganization": 1,
    "UpdatedWorkspacesStatuses": 1,
    "CreateOrUpdateGitHubPullRequest": 1,
    "TriggerInstantIntegration": 1,
    "PushCodebase": 1,
    "UpdatedGitHubPullRequest": 1,
  }
  analyze(folder, words)

```

	Word	Count	Search	Count	Markup	Ratio
0	Activity		43	1	43.000000	
1	Comment		94	3	31.333333	
2	LandWorkspaceChange		22	2	11.000000	
3	Members		19	2	9.500000	
4	AddPublicKey		7	1	7.000000	
5	ResolvedBy		6	1	6.000000	
6	Changes		6	1	6.000000	
7	UpdateInstallation		4	1	4.000000	
8	DownloadZip		4	1	4.000000	
9	CreateOrUpdateGitHubPullRequest		4	1	4.000000	
10	CreateComment		4	1	4.000000	
11	RemovePatches		7	2	3.500000	
12	VerifyEmail		7	2	3.500000	

13	UpdatedGitHubPullRequestStatuses	3	1	3.000000
14	UpdatedGitHubPullRequest	3	1	3.000000
15	SetupGitHubRepository	3	1	3.000000
16	CreateOrUpdateCodebaseRemote	3	1	3.000000
17	ReportWorkspacePresence	2	1	2.000000
18	UpdateCodebase	2	1	2.000000
19	DismissSuggestion	2	1	2.000000
20	UpdatedWorkspacesStatuses	2	1	2.000000
21	UpdateComment	2	1	2.000000
22	Notifications	2	1	2.000000
23	TriggerInstantIntegration	2	1	2.000000
24	Workspaces	4	2	2.000000
25	PushCodebase	2	1	2.000000
26	UpdateOrganization	2	1	2.000000
27	Integrations	2	1	2.000000
28	DeleteComment	2	1	2.000000
29	UpdateUser	4	2	2.000000
30	AddUserToCodebase	2	1	2.000000
31	ArchiveNotifications	2	1	2.000000
32	RequestReview	2	1	2.000000
33	WatchWorkspace	2	1	2.000000
34	DismissSuggestionHunks	2	1	2.000000
35	GenerateKeyPair	2	1	2.000000
36	UpdatedWorkspaceDiffs	2	1	2.000000
37	UpdatedView	2	1	2.000000
38	UpdatedSuggestion	2	1	2.000000
39	UpdatedOrganization	2	1	2.000000
40	LastUsedView	4	2	2.000000
41	DeleteIntegration	2	1	2.000000
42	PullCodebase	2	1	2.000000
43	UpdateACL	2	1	2.000000

	Word	Count	Search	Count	Markup	Ratio
0	MEAN		6.795455		1.204545	4.609848

```
[113]: folder = 'e:\\phd\\test_repos_graphql2\\toolkit-for-cardano\\'
words = {
    "WalletDelegate": 1,
    "WalletRegister": 1,
    "WalletFund": 1,
    "WalletCreate": 1,
    "SendFunds": 1,
    "TxSubmit": 1,
    "TxSign": 1,
    "TxBuild": 1,
    "Mint": 1,
    "Wallets": 1,
    "Utxos": 2,
```

```

    "TxFee": 1,
}
analyze(folder, words)

```

	Word	Count	Search	Count	Markup	Ratio
0	Utxos		22		2	11.0
1	Mint		9		1	9.0
2	WalletDelegate		2		1	2.0
3	WalletRegister		2		1	2.0
4	TxSubmit		2		1	2.0
5	TxSign		2		1	2.0
6	TxFee		2		1	2.0
7	WalletFund		1		1	1.0
8	WalletCreate		1		1	1.0
9	SendFunds		1		1	1.0
10	TxBuild		1		1	1.0
11	Wallets		1		1	1.0

	Word	Count	Search	Count	Markup	Ratio
0	MEAN		3.833333		1.083333	2.916667

```
[ ]:
```

```
[ ]:
```

```
[114]: global_array
```

```

[114]: array([2.50000000e+01, 2.10000000e+01, 2.10000000e+01, 1.10000000e+01,
              1.00000000e+01, 1.00000000e+01, 9.00000000e+00, 9.00000000e+00,
              9.00000000e+00, 9.00000000e+00, 4.00000000e+00, 1.00000000e+00,
              4.70000000e+01, 1.45000000e+01, 1.10000000e+01, 9.50000000e+00,
              8.50000000e+00, 8.00000000e+00, 6.00000000e+00, 6.00000000e+00,
              3.55000000e+01, 1.15000000e+01, 1.10000000e+01, 1.00000000e+01,
              8.50000000e+00, 8.00000000e+00, 8.00000000e+00, 8.00000000e+00,
              7.50000000e+00, 2.19000000e+02, 1.23000000e+02, 1.11000000e+02,
              3.50000000e+01, 2.70000000e+01, 2.60000000e+01, 1.20000000e+01,
              1.10000000e+01, 8.00000000e+00, 7.00000000e+00, 7.00000000e+00,
              7.00000000e+00, 6.00000000e+00, 5.00000000e+00, 5.00000000e+00,
              5.00000000e+00, 5.00000000e+00, 4.50000000e+00, 4.00000000e+00,
              4.00000000e+00, 4.00000000e+00, 4.00000000e+00, 4.00000000e+00,
              4.00000000e+00, 4.00000000e+00, 3.66666667e+00, 3.50000000e+00,
              3.50000000e+00, 3.50000000e+00, 2.00000000e+00, 2.00000000e+00,
              1.00000000e+00, 1.00000000e+00, 1.00000000e+00, 1.00000000e+00,
              1.00000000e+00, 1.00000000e+00, 1.00000000e+00, 1.00000000e+00,
              1.00000000e+00, 1.00000000e+00, 1.00000000e+00, 5.60000000e+01,
              3.20000000e+01, 2.60000000e+01, 8.00000000e+00, 7.00000000e+00,
              5.00000000e+00, 4.00000000e+00, 4.00000000e+00, 2.00000000e+00,
              2.00000000e+00, 2.00000000e+00, 2.00000000e+00, 1.00000000e+00,

```

1.00000000e+00, 1.00000000e+00, 1.00000000e+00, 1.00000000e+00,
1.00000000e+00, 1.00000000e+00, 1.00000000e+00, 1.00000000e+00,
1.00000000e+00, 1.00000000e+00, 1.00000000e+00, 1.48500000e+02,
1.00000000e+00, 1.00000000e+00, 1.00000000e+00, 1.00000000e+00,
1.00000000e+00, 2.57700000e+03, 6.60000000e+02, 3.79000000e+02,
3.71000000e+02, 2.70500000e+02, 1.80000000e+02, 1.64000000e+02,
1.49500000e+02, 1.43000000e+02, 1.22000000e+02, 9.90000000e+01,
8.20000000e+01, 6.60000000e+01, 6.10000000e+01, 5.50000000e+01,
5.40000000e+01, 5.40000000e+01, 4.10000000e+01, 3.90000000e+01,
3.70000000e+01, 3.30000000e+01, 3.10000000e+01, 2.70000000e+01,
2.60000000e+01, 2.10000000e+01, 1.70000000e+01, 1.20000000e+01,
1.20000000e+01, 1.20000000e+01, 1.00000000e+01, 1.00000000e+01,
9.50000000e+00, 9.00000000e+00, 9.00000000e+00, 8.00000000e+00,
8.00000000e+00, 8.00000000e+00, 8.00000000e+00, 7.00000000e+00,
7.00000000e+00, 7.00000000e+00, 4.00000000e+00, 4.00000000e+00,
3.00000000e+00, 6.36400000e+03, 1.44600000e+03, 1.09600000e+03,
1.04900000e+03, 7.70000000e+02, 7.12000000e+02, 4.42000000e+02,
4.26000000e+02, 4.20000000e+02, 3.87000000e+02, 3.45000000e+02,
3.41000000e+02, 3.19000000e+02, 2.53000000e+02, 2.46000000e+02,
2.19000000e+02, 1.88000000e+02, 1.82000000e+02, 1.69000000e+02,
1.68000000e+02, 1.56000000e+02, 1.48000000e+02, 1.46000000e+02,
1.31000000e+02, 1.05000000e+02, 8.20000000e+01, 8.20000000e+01,
7.30000000e+01, 7.20000000e+01, 6.00000000e+01, 5.90000000e+01,
5.10000000e+01, 5.10000000e+01, 4.70000000e+01, 3.20000000e+01,
2.30000000e+01, 2.30000000e+01, 1.80000000e+01, 1.70000000e+01,
1.70000000e+01, 1.70000000e+01, 1.70000000e+01, 1.60000000e+01,
1.60000000e+01, 1.40000000e+01, 1.20000000e+01, 1.20000000e+01,
9.00000000e+00, 8.00000000e+00, 4.00000000e+00, 2.00000000e+00,
2.00000000e+00, 2.00000000e+00, 2.00000000e+00, 2.00000000e+00,
2.00000000e+00, 2.00000000e+00, 2.00000000e+00, 8.50000000e+01,
7.50000000e+01, 5.40000000e+01, 3.75000000e+01, 3.60000000e+01,
2.60000000e+01, 5.00000000e+00, 3.50000000e+00, 3.00000000e+00,
3.00000000e+00, 2.00000000e+00, 2.00000000e+00, 1.00000000e+00,
1.00000000e+00, 1.00000000e+00, 1.00000000e+00, 1.00000000e+00,
1.00000000e+00, 1.00000000e+00, 1.00000000e+00, 1.00000000e+00,
1.00000000e+00, 1.00000000e+00, 1.00000000e+00, 1.00000000e+00,
1.00000000e+00, 1.00000000e+00, 3.00000000e+00, 1.00000000e+00,
1.00000000e+00, 1.00000000e+00, 1.09000000e+02, 9.50000000e+01,
5.40000000e+01, 4.40000000e+01, 4.40000000e+01, 7.00000000e+00,
4.63000000e+02, 3.53000000e+02, 1.80000000e+02, 5.50000000e+01,
7.00000000e+00, 6.00000000e+00, 4.00000000e+00, 4.00000000e+00,
3.50000000e+00, 3.50000000e+00, 3.50000000e+00, 3.00000000e+00,
3.00000000e+00, 3.00000000e+00, 3.00000000e+00, 2.50000000e+00,
2.50000000e+00, 1.10000000e+01, 1.10000000e+01, 1.00000000e+01,
1.00000000e+01, 7.00000000e+00, 7.00000000e+00, 6.00000000e+00,
5.00000000e+00, 4.30000000e+01, 3.13333333e+01, 1.10000000e+01,

```

9.50000000e+00, 7.00000000e+00, 6.00000000e+00, 6.00000000e+00,
4.00000000e+00, 4.00000000e+00, 4.00000000e+00, 4.00000000e+00,
3.50000000e+00, 3.50000000e+00, 3.00000000e+00, 3.00000000e+00,
3.00000000e+00, 3.00000000e+00, 2.00000000e+00, 2.00000000e+00,
2.00000000e+00, 2.00000000e+00, 2.00000000e+00, 2.00000000e+00,
2.00000000e+00, 2.00000000e+00, 2.00000000e+00, 2.00000000e+00,
2.00000000e+00, 2.00000000e+00, 2.00000000e+00, 2.00000000e+00,
2.00000000e+00, 2.00000000e+00, 2.00000000e+00, 2.00000000e+00,
2.00000000e+00, 2.00000000e+00, 2.00000000e+00, 2.00000000e+00,
2.00000000e+00, 2.00000000e+00, 2.00000000e+00, 2.00000000e+00,
2.00000000e+00, 1.10000000e+01, 9.00000000e+00, 2.00000000e+00,
2.00000000e+00, 2.00000000e+00, 2.00000000e+00, 2.00000000e+00,
1.00000000e+00, 1.00000000e+00, 1.00000000e+00, 1.00000000e+00,
1.00000000e+00])

```

```

[135]: data = global_array
#
bins = [1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 15, 20, 30, 50, 100, 7000]

#
intervals = [
    '1', '2', '3', '4', '5', '6', '7', '8', '9', '10',
    '11-15', '16-20', '21-30', '31-50', '51-100', '>100'
]

#
counts, _ = np.histogram(data, bins=bins)

#           {           }
result = dict(zip(intervals, counts))
print(result)

```

```

{'1': 56, '2': 50, '3': 22, '4': 20, '5': 7, '6': 7, '7': 13, '8': 13, '9': 11,
'10': 24, '11-15': 8, '16-20': 12, '21-30': 17, '31-50': 22, '51-100': 43,
'>100': 0}

```

```

[150]: print(len(data[data==1]))
print(len(data[(data>1)&(data<=4)]))
print(len(data[(data>5)&(data<=10)]))
print(len(data[(data>10)]))

```

```

56
91
51
119

```

```

[ ]:

```